Grade 7 | Name______________________
Unit 4 | Date______________________
Lesson 5 Practice | Period____________________

For questions 1-3, complete each statement with the correct vocabulary from the box below.

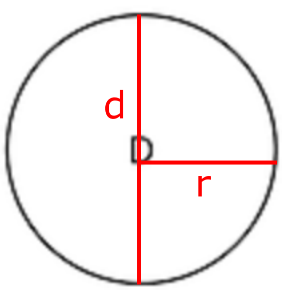
Radius	Diameter	Area	Circumference

1. The ____radius____ is a line segment extending from the center of a circle to a point on the circle.

2. The ____circumference____ measures the distance around the circle.

3. The ____diameter____ is a line segment from any point on the circle passing through the center to another point on the circle.

4. On circle D, sketch and label the radius, diameter, and circumference.



C (measurement around)

r = radius
d = diameter
c = circumference

5. The following objects were measured. Complete the table below to determine a relationship between the circumference and the diameter. Round to the nearest hundredth.

Object	Diameter (in)	Circumference (in)	$C \div D$
Clock	11.1	34.88	$34.88 \div 11.1 = 3.14$
Masking Tape	5.75	18	$18 \div 5.75 = 3.13$

6. For any circular object, the circumference divided by the diameter is approximately ____3.14____.

7. Jan measured the diameter and circumference of several circular objects and recorded his measurements in the table. Using your estimate of $C \div D$, determine which measurement is inaccurate.

Object	Diameter (cm)	Circumference (cm)
Ring	2	6 $\frac{6}{2} = 3$
Top of a mug	7	23 $\frac{23}{7} = 3.286$
Pizza	31	63 $\frac{63}{31} = 2.032$
A coin	3	10 $\frac{10}{3} = 3.\overline{3}$

8. Select all the pairs that could be reasonable approximations for the diameter and circumference of a circle. Explain and show your reasoning.

A. 7 meters and 23 meters.
≈ 3.286 $\frac{23}{7} =$ approximately 3.286, close to 3.14

B. 15 inches and 63 inches.
≈ 4.2 $\frac{63}{15} =$ approximately 4.2

C. 27 centimeters and 68 centimeters.
≈ 2.519 $\frac{68}{27} =$ approximately 2.519

For questions 9-10, use your approximation of $C \div D$ to find either the diameter or circumference.

9. If the diameter of a magnifying glass is 5.2 centimeters, find the approximate circumference.
$C = \pi d$
$C = 3.14 \cdot 5.2$
$C = 16.328$
$C \approx 16.328$ cm

10. If a car tire has a circumference of 71.6 feet, find the approximate diameter to the nearest tenth.
$\frac{C}{\pi} = \frac{\pi d}{\pi}$
$\frac{C}{\pi} = d$
$\frac{71.6}{3.14} = d$
≈ 22.8 ft